

MediTrack — Clinical Quality & Readmission Analytics

Hospital-Wide Quality Assurance, Department Benchmarking & Decision Support Audit Report

99,340	11.39%	0.18 (Tuned)	4.38 Days	\$3.08M / 10k
Evaluated Encounters	30d Readmission Base Rate	Clinical Decision Threshold	Avg Length of Stay	Simulated Net Savings

1. Department & Medical Specialty Readmission Performance

Clinical departments with the highest encounter volumes and corresponding 30-day readmission rates. Units exceeding the 11.4% hospital average are prioritized for dedicated discharge nurse transitional outreach.

Department / Specialty	Encounters	Avg Stay (Days)	30d Readmissions	Readmission Rate
General Inpatient / Unspecified	48,614	4.4	5,753	11.8%
InternalMedicine	14,237	4.6	1,642	11.5%
Emergency/Trauma	7,419	4.4	845	11.4%
Family/GeneralPractice	7,252	4.4	879	12.1%
Cardiology	5,278	3.5	424	8.0%
Surgery-General	3,059	4.5	342	11.2%
Nephrology	1,539	5.0	248	16.1%
Orthopedics	1,392	4.0	151	10.8%
Orthopedics-Reconstructive	1,230	3.9	92	7.5%
Radiologist	1,121	3.5	102	9.1%

2. Primary Diagnosis Categories & Length of Stay (LOS) Distribution

Primary Diagnosis Category	Encounters	Avg Stay (Days)	Readmissions	30d Readm %
Circulatory	29,680	4.2	3,469	11.7%
Other	17,793	4.8	2,079	11.7%
Respiratory	13,934	4.2	1,402	10.1%
Digestive	9,333	4.4	1,010	10.8%
Diabetes	8,661	4.3	1,135	13.1%
Injury	6,851	4.6	850	12.4%
Genitourinary	5,002	4.2	552	11.0%
Musculoskeletal	4,935	3.9	471	9.5%
Neoplasms	3,131	5.3	341	10.9%
Missing_or_Other	20	3.8	5	25.0%

3. High-Risk Cohort Drill-Through (Care Team Triage Worklist)

Multivariate cohort stratification cross-referencing age bracket, gender, and prior inpatient admissions. Patients with repeated prior inpatient hospitalizations exhibit acute readmission risk spikes approaching 30-45%.

Age Group	Gender	Prior Inpatient Tier	Cohort Size	30d Readmissions	Readmission Rate
[20-30)	Female	2+ Prior Inpatient	254	103	40.6%
[30-40)	Female	2+ Prior Inpatient	362	120	33.1%
[30-40)	Male	2+ Prior Inpatient	277	67	24.2%
[40-50)	Male	2+ Prior Inpatient	680	163	24.0%
[40-50)	Female	2+ Prior Inpatient	789	188	23.8%
[50-60)	Female	2+ Prior Inpatient	1,190	274	23.0%
[60-70)	Male	2+ Prior Inpatient	1,488	335	22.5%
[70-80)	Male	2+ Prior Inpatient	1,639	351	21.4%

4. Clinical Decision Support & Guideline Implementation Audit

AHRQ Project RED Compliance: All patients with risk scores exceeding the operational threshold (0.18) automatically receive a structured After-Hospital Care Plan (AHOP) and 48-hour phone call priority.

ADA Glycemic Transitions: Inpatients on insulin therapy or with acute dosage changes are scheduled for ambulatory medication reconciliation within 7-14 days with hypoglycemia teach-back education.

Safety Guardrail Enforcement: The embedded assistant strictly blocks diagnostic and prescriptive inquiries, preserving clinical accountability and directing all diagnostic/dosing authority to licensed attending physicians.